F70LA Life Insurance Mathematics A (2025/26)
Tutorial 5 Solutions

1. The insurer will be more rigorous with underwriting for term insurance than for whole life insurance because the potential financial consequence of adverse selection is greater.

   Note that the insurer expects few claims to arise from the term insurance portfolio. Premiums are small, relative to the death benefit, because the probability of payment of the death benefit is assumed to be small.

   For whole life insurance, premiums are substantially larger, as payment of the death benefit is a certain event (ignoring surrenders). The only uncertainty is the timing of the benefit payment.

   The main risk to the insurer is that a life with a very high mortality risk, much higher than assumed for the insured population, purchases life insurance. It is likely in this case that the life will pay very few premiums, and the policy will involve a large death benefit payout with very little premium income. Since term insurance has much lower premiums for a given sum insured than whole life insurance, it is likely that such a policyholder would choose term insurance.

   Hence, the risk of adverse selection is greater for term insurance than for whole life insurance, and underwriting is used to reduce the adverse selection risk.

2. (a) Without term insurance, the homeowner's dependents may struggle to meet mortgage payments in the event of the homeowner's death. The lending company wishes to reduce as far as possible the risk of having to foreclose on the loan, i.e. take possession of the house and evict the occupants in order the sell house. Foreclosure is expensive, creates hardship for the homeowner's family at the worst possible time and, in some jurisdictions, may be rarely allowed by the courts.

   Term insurance is used to pay off the mortgage balance in the event of the homeowner's death, thus avoiding the foreclosure risk for both the lender and homeowner's family.

   (b) If the homeowner is paying regular instalments of capital and interest to pay off the mortgage, then the term insurance sum insured will decrease as the loan outstanding decreases. The reduction in loan outstanding is slow in the early years of, say, a 25-year mortgage, but speeds up later. The reduction in the term insurance sum insured is therefore not linear.

   Different loan provisions, including interest-only loan periods, fixed or variable interest rates and fixed or variable repayment instalments, will all affect the sum insured.

   (c) Around 2%-3% of applicants for insurance are considered to be too high risk. If these lives are, in consequence, unable to purchase property, then that is a social cost which may not be acceptable to society.

3. Insurers prefer policies to remain *in force*, i.e. premiums are being paid as required under the contractual terms of the insurance policy and so the insurance benefits will be paid if the specified insurance events (e.g. the death of the insured life) occur. An insurer's profits from long-term business arise largely from the difference between the interest earned on the accumulated premiums and the interest needed to support the benefits (called *the interest spread*).

   After age 80, few policyholders will be receiving any salary, so there is greater risk that the premiums will not be affordable. Policyholders may then surrender their policies, cutting off the profit stream to the insurer.

   By designing the contract such that no premiums are due after age 80, the insurer increases significantly the proportion of policies that remain in force at that time, which we call *the persistency*.

x	ℓ_x	A_x
55	99 176.23	0.29962
56	99 072.49	0.31088
57	98 953.70	0.32250
58	98 817.64	0.33450
59	98 662.00	0.34685
60	98 484.31	0.35954

4. (a) With $v := 1.04^{-1}$,

$$ {}_5|A_{55} = v^5\, {}_5p_{55}\, A_{60} = 1.04^{-5}\,\frac{98\,484.31}{99\,176.23}\,(0.35954) = 1.04^{-5}(0.99302)(0.35954) = 0.2934540 $$

   and so $100\,000\,{}_5|A_{55} = £29\,345.40$.

   (b)

$$ \begin{aligned} {}_3|A^1_{55:\overline{2}|} = v^3\, {}_3p_{55}\, A^1_{58:\overline{2}|} &= v^3\, {}_3p_{55}\, \left(A_{58} - v^2\, {}_2p_{58}\, A_{60}\right) \\ &= 1.04^{-3}\frac{98\,817.64}{99\,176.23}\,\left(0.33450 - 1.04^{-2}\,\frac{98\,484.31}{98\,817.64}\,(0.35954)\right) \\ &= 1.04^{-3}(0.996384)\,\left(0.33450 - 1.04^{-2}\,(0.99663)\,(0.35954)\right) \\ &= 0.0028401 \end{aligned} $$

   and so $100\,000\,{}_3|A^1_{55:\overline{2}|} = £284.01$.

   (c)

$$ {}_3|A_{55:\overline{2}|} = v^3\, {}_3p_{55}\, A_{58:\overline{2}|} = v^3\, {}_3p_{55}\, \left(A^1_{58:\overline{2}|} + A_{58:\overline{2}|}^{\,1}\right) = {}_3|A^1_{55:\overline{2}|} + v^5\, {}_5p_{55}. $$

   From part (b), ${}_3|A^1_{55:\overline{2}|} = 0.0028401$ and

$$ v^5\, {}_5p_{55} = 1.04^{-5}\frac{98\,484.31}{99\,176.23} = 0.8161928. $$

   Hence

$$ 100\,000\,{}_3|A_{55:\overline{2}|} = 284.01 + 81\,619.28 = £81\,903.29. $$

5. $A_x^{(m)}$ is the EPV of a payment of 1 unit if death occurs, with the payment made at the end of the $m$thly period in which death occurs. We want to write an expression for the EPV so that we can apply the UDD assumption.

   Fix a positive integer k and consider the present value of the $m$thly-paid death benefit over the ages $[x+k, x+k+1)$. To calculate it, divide the year into m sub-intervals, each of length $1/m$, i.e. $[x+k, x+k+\frac{1}{m})$, $[x+k+\frac{1}{m}, x+k+\frac{2}{m})$, ..., $[x+k+\frac{m-1}{m}, x+k+1)$.

   The present value of the $m$thly-paid death benefit over the ages $[x+k, x+k+1)$ is

$$\sum_{s=0}^{m-1} v^{k+\frac{s+1}{m}} \; {}_{k+\frac{s}{m}}p_x \; {}_{\frac{1}{m}}q_{x+k+\frac{s}{m}} = v^k \, {}_kp_x \sum_{s=0}^{m-1} v^{\frac{s+1}{m}} \; {}_{\frac{s}{m}}p_{x+k} \; {}_{\frac{1}{m}}q_{x+k+\frac{s}{m}}, \tag{1}$$

   where the right-hand side follows from ${}_{k+\frac{s}{m}}p_x = {}_kp_x \, {}_{\frac{s}{m}}p_{x+k}$. Then $A_x^{(m)}$ will be the sum of the present values over all possible values of k, i.e.

$$A_x^{(m)} = \sum_{k=0}^{\infty} v^k \, {}_kp_x \sum_{s=0}^{m-1} v^{\frac{s+1}{m}} \; {}_{\frac{s}{m}}p_{x+k} \; {}_{\frac{1}{m}}q_{x+k+\frac{s}{m}}.$$

   Next re-write the fractional age probabilities so that the UDD assumption can be applied. Using ${}_{\frac{1}{m}}p_{x+k+\frac{s}{m}} = 1 - {}_{\frac{1}{m}}q_{x+k+\frac{s}{m}}$,

$$\begin{aligned} {}_{\frac{s}{m}}p_{x+k} \; {}_{\frac{1}{m}}q_{x+k+\frac{s}{m}} = {}_{\frac{s}{m}}p_{x+k} \left(1 - {}_{\frac{1}{m}}p_{x+k+\frac{s}{m}}\right) &= {}_{\frac{s}{m}}p_{x+k} - {}_{\frac{s}{m}+\frac{1}{m}}p_{x+k} \\ &= \left(1 - {}_{\frac{s}{m}}q_{x+k}\right) - \left(1 - {}_{\frac{s+1}{m}}q_{x+k}\right) \\ &= {}_{\frac{s+1}{m}}q_{x+k} - {}_{\frac{s}{m}}q_{x+k}. \end{aligned}$$

   The UDD assumption over integer ages, i.e. for $u \in [0,1)$ and integer x, ${}_uq_x = uq_x$, gives

$${}_{\frac{s+1}{m}}q_{x+k} - {}_{\frac{s}{m}}q_{x+k} = \frac{s+1}{m}q_{x+k} - \frac{s}{m}q_{x+k} = \frac{1}{m}q_{x+k}.$$

   Substituting the UDD assumption into the expression on the right-hand side of (1),

$$\sum_{s=0}^{m-1} v^{k+\frac{s+1}{m}} \; {}_{k+\frac{s}{m}}p_x \; {}_{\frac{1}{m}}q_{x+k+\frac{s}{m}} = v^k \, {}_kp_x q_{x+k} \, \frac{1}{m} \sum_{s=0}^{m-1} v^{\frac{s+1}{m}}.$$

   Now consider in isolation the finite geometric sum and recalling that $(1+i^{(m)}/m)^m = 1+i$,

$$\sum_{s=0}^{m-1} v^{\frac{s+1}{m}} = \frac{1-v}{(1+i)^{1/m}-1} = \frac{iv}{i^{(m)}/m} = mv\frac{i}{i^{(m)}}.$$

   Hence

$$\sum_{s=0}^{m-1} v^{k+\frac{s+1}{m}} \; {}_{k+\frac{s}{m}}p_x \; {}_{\frac{1}{m}}q_{x+k+\frac{s}{m}} = v^k \, {}_kp_x q_{x+k} \, \frac{1}{m} \, mv\frac{i}{i^{(m)}} = v^{k+1} \, {}_kp_x q_{x+k} \, \frac{i}{i^{(m)}}.$$

Summing the last expression over all possible values of k,

$$A_x^{(m)} = \sum_{k=0}^{\infty} v^k {}_kp_x q_{x+k} \frac{1}{m} mv \frac{i}{i^{(m)}} = \sum_{k=0}^{\infty} v^{k+1} {}_kp_x q_{x+k} \frac{i}{i^{(m)}} = \frac{i}{i^{(m)}} A_x.$$

6. The EPV $A_x^{(m)}$ is

$$A_x^{(m)} = \mathbb{E}\left(v^{K_x^{(m)}+\frac{1}{m}}\right) = \sum_{k=0}^{\infty} v^{\frac{k+1}{m}} {}_{\frac{k}{m}}p_x \, {}_{\frac{1}{m}}q_{x+\frac{k}{m}}$$

Expanding the summation,

$$\begin{aligned} A_x^{(m)} =& v^{\frac{1}{m}} {}_{\frac{1}{m}}q_x + v^{\frac{2}{m}} {}_{\frac{1}{m}}p_x \, {}_{\frac{1}{m}}q_{x+\frac{1}{m}} + v^{\frac{3}{m}} {}_{\frac{2}{m}}p_x \, {}_{\frac{1}{m}}q_{x+\frac{2}{m}} + \cdots \\ =& v^{\frac{1}{m}} {}_{\frac{1}{m}}q_x + v^{\frac{1}{m}} {}_{\frac{1}{m}}p_x \left(v^{\frac{1}{m}} {}_{\frac{1}{m}}q_{x+\frac{1}{m}} + v^{\frac{2}{m}} {}_{\frac{1}{m}}p_{x+\frac{1}{m}} \, {}_{\frac{1}{m}}q_{x+\frac{2}{m}} + \cdots\right) \\ =& v^{\frac{1}{m}} {}_{\frac{1}{m}}q_x + v^{\frac{1}{m}} {}_{\frac{1}{m}}p_x \left(A_{x+\frac{1}{m}}^{(m)}\right). \end{aligned}$$

7. We need to calculate $\bar{A}_{40:\overline{20|}} = \bar{A}^1_{40:\overline{20|}} + v^{20} {}_{20}p_{40}$. So we need to determine both $v^{20} {}_{20}p_{40}$ and $\bar{A}^1_{40:\overline{20|}}$. First, calculate $v^{20} {}_{20}p_{40}$. We know

$$A^1_{40:\overline{20|}} = A_{40:\overline{20|}} - v^{20} {}_{20}p_{40} \qquad \text{and} \qquad A^1_{40:\overline{20|}} = A_{40} - v^{20} {}_{20}p_{40} A_{60},$$

thus

$$A_{40:\overline{20|}} - v^{20} {}_{20}p_{40} = A_{40} - v^{20} {}_{20}p_{40} A_{60} \quad \Rightarrow v^{20} {}_{20}p_{40} = \frac{A_{40:\overline{20|}} - A_{40}}{1 - A_{60}} = \frac{0.55 - 0.25}{1 - 0.4} = 0.5.$$

Now to calculate $\bar{A}^1_{40:\overline{20|}}$ using each of the required approximations. Note that

$$A^1_{40:\overline{20|}} = A_{40:\overline{20|}} - v^{20} {}_{20}p_{40} = 0.55 - 0.5 = 0.05.$$

(a) Under claims acceleration, the relevant approximation is $\bar{A}_x \approx (1+i)^{0.5} A_x$. Under claims accceleration,

$$\bar{A}^1_{40:\overline{20|}} = \bar{A}_{40} - v^{20} {}_{20}p_{40} \bar{A}_{60} \approx (1+i)^{0.5} A_{40} - v^{20} {}_{20}p_{40} \left((1+i)^{0.5} A_{60}\right) = (1+i)^{0.5} A^1_{40:\overline{20|}}.$$

Thus, under claims accceleration,

$$100\,000 \bar{A}_{40:\overline{20|}} \approx 100\,000 \left((1+i)^{0.5} A^1_{40:\overline{20|}} + v^{20} {}_{20}p_{40}\right) = 100\,000 \left((1.06)^{0.5} 0.05 + 0.5\right) = 55\,147.82.$$

(b) Under UDD, the relevant approximation is $\bar{A}_x \approx \frac{i}{\delta} A_x$. This means that, under UDD,

$$\bar{A}^1_{40:\overline{20|}} = \bar{A}_{40} - v^{20} {}_{20}p_{40} \bar{A}_{60} \approx \frac{i}{\delta} A_{40} - v^{20} {}_{20}p_{40} \left(\frac{i}{\delta} A_{60}\right) = \frac{i}{\delta} A^1_{40:\overline{20|}}.$$

Thus, under UDD,

$$100\,000 \bar{A}_{40:\overline{20|}} \approx 100\,000 \left(\frac{i}{\delta} A^1_{40:\overline{20|}} + v^{20} {}_{20}p_{40}\right) = 100\,000 \left(\frac{1.06}{\ln(1.06)} 0.05 + 0.5\right) = 55\,148.54.$$

x	$\ell_{[x]}$	$\ell_{[x]+1}$	$\ell_{[x]+2}$	$\ell_{[x]+3}$	ℓ_{x+4}	$x+4$
40	100 000	99 899	99 724	99 520	99 288	44
41	99 802	99 689	99 502	99 283	99 033	45
42	99 597	99 471	99 268	99 030	98 752	46
43	99 365	99 225	99 007	98 747	98 435	47
44	99 120	98 964	98 726	98 429	98 067	48

8. Set $v := 1.05^{-1}$. We want to calculate $100\,000\,{}_{1|}A^{1}_{[40]:\overline{4}|}$. First,

$$\begin{aligned}
{}_{1|}A^{1}_{[40]:\overline{4}|} =& v\, p_{[40]}\, A^{\ 1}_{[40]+1:\overline{4}|} \\
=& v\, p_{[40]} \left(v\, q_{[40]+1} + v^2\, p_{[40]+1}\, q_{[40]+2} + v^3\, {}_2p_{[40]+1}\, q_{[40]+3} + v^4\, {}_3p_{[40]+1}\, q_{44}\right) \\
=& v^2\, p_{[40]}\, q_{[40]+1} + v^3\, {}_2p_{[40]}\, q_{[40]+2} + v^4\, {}_3p_{[40]}\, q_{[40]+3} + v^5\, {}_4p_{[40]}\, q_{44} \\
=& v^2\, \frac{\ell_{[40]+1}}{\ell_{[40]}}\, \frac{\ell_{[40]+1} - \ell_{[40]+2}}{\ell_{[40]+1}} + v^3\, \frac{\ell_{[40]+2}}{\ell_{[40]}}\, \frac{\ell_{[40]+2} - \ell_{[40]+3}}{\ell_{[40]+2}} + v^4\, \frac{\ell_{[40]+3}}{\ell_{[40]}}\, \frac{\ell_{[40]+3} - \ell_{44}}{\ell_{[40]+3}} \\
& \qquad + v^5\, \frac{\ell_{44}}{\ell_{[40]}}\, \frac{\ell_{44} - \ell_{45}}{\ell_{44}} \\
=& v^2\, \frac{\ell_{[40]+1} - \ell_{[40]+2}}{\ell_{[40]}} + v^3\, \frac{\ell_{[40]+2} - \ell_{[40]+3}}{\ell_{[40]}} + v^4\, \frac{\ell_{[40]+3} - \ell_{44}}{\ell_{[40]}} + v^5\, \frac{\ell_{44} - \ell_{45}}{\ell_{[40]}} \\
=& v^2\, \frac{99\,899 - 99\,724}{100\,000} + v^3\, \frac{99\,724 - 99\,520}{100\,000} + v^4\, \frac{99\,520 - 99\,288}{100\,000} + v^5\, \frac{99\,288 - 99\,033}{100\,000} \\
=& \frac{v^2\left(175 + 204v + 232v^2 + 255v^3\right)}{100\,000} \\
=& 0.00725619.
\end{aligned}$$

Then $100\,000\,{}_{1|}A^{1}_{[40]:\overline{4}|} = £725.62$.

9. Let the random variable K_x represent the curtate future lifetime of (x). Then

$$Z_1 = \begin{cases} v^{K_x+1} & \text{if } K_x \le n-1, \\ 0 & \text{if } K_x \ge n. \end{cases}$$

and

$$Z_2 = v^{K_x+1}.$$

Taking expectations,

$$\mathbb{E}(Z_1) = A^{1}_{x:\overline{n}|} \qquad \text{and} \qquad \mathbb{E}(Z_2) = A_x.$$

The covariance is

$$\mathbb{C}ov(Z_1, Z_2) = \mathbb{E}\left(Z_1\, Z_2\right) - \mathbb{E}\left(Z_1\right)\mathbb{E}\left(Z_2\right).$$

As

$$Z_1\,Z_2 = \begin{cases} v^{K_x+1} \times v^{K_x+1} = v^{2K_x+2} = \left(v^2\right)^{K_x+1} & \text{if } K_x \leq n-1, \\ 0 \times v^{K_x+1} = 0 & \text{if } K_x \geq n. \end{cases}$$

the present value of $Z_1\,Z_2$ is an n-year term insurance with sum insured £1 but valued using the interest rate $(1+i)^2 - 1$. Taking expectations gives

$$\mathbb{E}\left(Z_1\,Z_2\right) = {}^2A^1_{x:\overline{n}|}.$$

Thus

$$\mathbb{C}ov(Z_1, Z_2) = {}^2A^1_{x:\overline{n}|} - A^1_{x:\overline{n}|}\,A_x.$$

10. (a) (i) It pays:
    - 50 units immediately upon (x) dying in the first 20 years.

    (ii) It pays:
    - 10 units immediately at time 5, if (x) survives to that time, and
    - 20 units immediately upon (x) dying, if (x) dies after 5 years.

    (b)

$$\mathbb{E}(Z_1) = 50 \int_0^{20} v^t\,{}_tp_x\,\mu_{x+t}\,dt$$

and

$$\mathbb{E}(Z_2) = 10\,v^5\,{}_5p_x + 20 \int_5^{\infty} v^t\,{}_tp_x\,\mu_{x+t}\,dt.$$

(c) Any of the expressions below are correct.

$$\mathbb{E}(Z_1) = 50\,\bar{A}^1_{x:\overline{20}|}.$$

and

$$\mathbb{E}(Z_2) = 10\,v^5\,{}_5p_x + 20\,{}_{5|}\bar{A}_x = 10\,{}_5E_x + 20\,{}_{5|}\bar{A}_x = 10\,A_{x:\overline{5}|}^{\;\;\frac{1}{}} + 20\,{}_{5|}\bar{A}_x$$

(d) The covariance is

$$\mathbb{C}ov(Z_1, Z_2) = \mathbb{E}\left(Z_1\,Z_2\right) - \mathbb{E}\left(Z_1\right)\mathbb{E}\left(Z_2\right).$$

We've already calculated $\mathbb{E}\left(Z_1\right)$ and $\mathbb{E}\left(Z_2\right)$, so it remains to calculate $\mathbb{E}\left(Z_1 Z_2\right)$. First,

$$Z_1\,Z_2 = \begin{cases} 50v^{T_x} \times 0 = 0 & \text{if } T_x \in (0,5], \\ 50v^{T_x} \times \left(10v^5 + 20v^{T_x}\right) = 500v^{5+T_x} + 1000v^{2T_x} & \text{if } T_x \in (5,20], \\ 0 \times \left(10v^5 + 20v^{T_x}\right) = 0 & \text{if } T_x > 20. \end{cases}$$

Then

$$\begin{aligned} \mathbb{E}\left(Z_1 Z_2\right) &= \int_5^{20} \left(500v^{5+t} + 1000v^{2t}\right)\,{}_tp_x\,\mu_{x+t}\,dt \\ &= 500\,v^5 \int_5^{20} v^t\,{}_tp_x\,\mu_{x+t}\,dt + 1000 \int_5^{20} v^{2t}\,{}_tp_x\,\mu_{x+t}\,dt \\ &= 500\,v^5\,{}_{5|}\bar{A}^1_{x:\overline{15}|} + 1000\,{}^2_{5|}\bar{A}^1_{x:\overline{15}|}. \end{aligned}$$

Hence

$$\begin{aligned}\mathbb{C}ov(Z_1, Z_2) =& 500\, v^5\, {}_{5|}\bar{A}^1_{x:\overline{15|}} + 1000\, {}^2_{5|}\bar{A}^1_{x:\overline{15|}} - \left(50\, \bar{A}^1_{x:\overline{20|}}\right)\left(10\, v^5\, {}_5p_x + 20\, {}_{5|}\bar{A}_x\right)\\ & 500\, v^5\, {}_{5|}\bar{A}^1_{x:\overline{15|}} + 1000\, {}^2_{5|}\bar{A}^1_{x:\overline{15|}} - 500\, \bar{A}^1_{x:\overline{20|}}\, v^5\, {}_5p_x - 1000\, \bar{A}^1_{x:\overline{20|}}\, {}_{5|}\bar{A}_x.\end{aligned}$$

11. (a)

$$\begin{aligned}(IA)^1_{x:\overline{n|}} =& \sum_{k=0}^{n-1}(k+1)v^{k+1}\, {}_k|q_x\\ =& v\, q_x + 2v^2\, p_x\, q_{x+1} + 3v^3\, {}_2p_x\, q_{x+2} + 4v^4\, {}_3p_x\, q_{x+3} + \cdots + nv^n\, {}_{n-1}p_x\, q_{x+n-1}\\ =& v\, q_x + v\, p_x\left(2v\, q_{x+1} + 3v^2\, p_{x+1}\, q_{x+2} + 4v^3\, {}_2p_{x+1}\, q_{x+3} + \cdots + nv^{n-1}\, {}_{n-2}p_{x+1}\, q_{x+n-1}\right)\\ =& v\, q_x + v\, p_x\Bigg((1+1)v\, q_{x+1} + (1+2)v^2\, p_{x+1}\, q_{x+2} + (1+3)v^3\, {}_2p_{x+1}\, q_{x+3}\\ & \qquad + \cdots + (1+n-1)v^{n-1}\, {}_{n-2}p_{x+1}\, q_{x+n-1}\Bigg)\\ =& v\, q_x + v\, p_x\Bigg(v\, q_{x+1} + v^2\, p_{x+1}\, q_{x+2} + v^3\, {}_2p_{x+1}\, q_{x+3} + \cdots + v^{n-1}\, {}_{n-2}p_{x+1}\, q_{x+n-1}\\ & \qquad + v\, q_{x+1} + 2v^2\, p_{x+1}\, q_{x+2} + 3v^3\, {}_2p_{x+1}\, q_{x+3} + \cdots + (n-1)v^{n-1}\, {}_{n-2}p_{x+1}\, q_{x+n-1}\Bigg)\\ =& vq_x + vp_x\left((IA)^1_{x+1:\overline{n-1|}} + A^1_{x+1:\overline{n-1|}}\right).\end{aligned}$$

(b) Since

$$\lim_{n\to\infty}(IA)^1_{x:\overline{n|}} = (IA)_x \qquad \text{and} \qquad \lim_{n\to\infty} A^1_{x:\overline{n|}} = A_x,$$

letting $n \to \infty$ in the relationship in part (a) gives

$$(IA)_x = vq_x + vp_x\left((IA)_{x+1} + A_{x+1}\right).$$

(c) We are given that, an annual effective rate of 6% per annum, $(IA)_{50} = 4.99675$, $A_{51} = 0.24905$ and $A^1_{50:\overline{1|}} = 0.00558$.
Using the result in part (b), with $x = 50$,

$$(IA)_{50} = vq_{50} + vp_{50}\left((IA)_{51} + A_{51}\right) \qquad \Rightarrow \qquad (IA)_{51} = \frac{(IA)_{50} - vq_{50}}{vp_{50}} - A_{51}.$$

To determine the probability q_{50}, consider

$$A^1_{50:\overline{1|}} = vq_{50} \qquad \Rightarrow \qquad q_{50} = (1+i)A^1_{50:\overline{1|}} = 1.06 \times 0.00558 = 0.0059148.$$

Hence

$$(IA)_{51} = \frac{(IA)_{50} - vq_{50}}{vp_{50}} - A_{51} = \frac{4.99675 - 0.00558}{1.06^{-1}(1 - 0.0059148)} - 0.24905 = 5.0731.$$